

# Second Challenge

Autonomous Software Agents

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## Requirements for the participation

- Each group will have **two agents**: one BDI and one LLM.
- **Standard mission:**
  - Collect and deliver parcels (as in the first challenge)
  - Both agents (BDI and LLM) can perform this mission at the same time
- **Special missions**
  - DeliverooJS will also send you special missions written in natural language.
  - These missions must be: [read by the LLM agent](#), [interpreted by the LLM agent](#), [executed by your system](#). **They require an LLM.**
  - [Examples of special missions are shown in the following slides](#). They are very similar to the ones you will encounter during the challenge
  - Special missions usually give **many more points** than standard parcel collection and delivery.
  - However, some special missions **may not always be worth completing**.
  - Your system should decide when it is convenient to complete a special mission and when it is better to ignore it.

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# Special missions - Examples

Special missions are divided into three levels of complexity:

**1. Atomic special missions:**

- Relatively simple, atomic missions
- Can be solved with small modifications to the code shown in the LLM-based agents tutorials

**2. Intermediate special missions:**

- Require additional internal tools to be completed efficiently
- Are non-atomic but persistent: they remain active for the entire duration of the match
- Require the agent to adapt its game strategy in order to satisfy the prompt request.

**3. Missions requiring coordination or communication**

- Require a communication mechanism between:
  - the BDI agent and the LLM agent
  - The LLM agent and the game chat

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## Special missions – Level 1

Atomic missions require standard tools to be completed. E.g., move, calculate, pick-up, put-down, ...

Examples:

- Move to coordinate (4,7) and you get **+10pts**
  - Move to  $x=4*2$   $y=(1+3)*3$  to get **-10pts**
  - Drop a package in the leftmost tile to get **5pt**
  - Drop a package in the leftmost tile to get **-10pt**
  - What is the capital of Italy?
  - Calculate  $5*5$
- } Send the answer to the agent who sent the prompt

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## Special missions – Level 2

Level 2 missions require the agent to adapt its game strategy in order to satisfy the prompt request. Consider using additional tools to modify the standard pick-up and delivery behaviour

Examples:

- Deliver stacks of exactly **3** parcels at a time to **double** the reward
- Deliver stacks of exactly **5** parcels at a time to get **0.3** of the standard reward
- Every time you deliver in **(x1,y1) or (x2,y2)** you get **5x pts** than in a regular delivery tile
- Every time you deliver in **(x1,y1)** you get **0 pts**
- If you deliver parcels with a score higher than 10, **you get no reward.**
- Do not go through tile **(x,y) otherwise you lose 50pts.**

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## Special missions – Level 3

Level 3 missions require multi-agent coordination and communication tools.

Examples:

- Move both agents to the neighborhood of position **(x,y)** within a maximum distance of **3**, and have them wait for each other. You will receive **500pts.**
- If a parcel is initially picked up by one agent and later delivered by the other agent, you will receive a **200 points bonus.**
- All agents must move to an odd-numbered row and wait for our message before moving again, as in a “red light, green light” game. **700 points bonus.**

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# Requirements for the participation

- To participate in the second challenge, register your agent by filling out the following form: <https://forms.gle/i1D7c6Fycpdi2mwi6> **DEADLINE: 27/05**
- **Remember**
  - Your agents do not need to be perfect to participate in the challenge.
  - Even with a BDI agent and an LLM agent that can solve only Level 1 tasks, you can still earn points.
  - The challenge is just a game and an opportunity to informally test your agents together.
  - It does **not** contribute to your final grade.

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## Remaining lessons

- 20/05 → Exam presentation and technical support challenge 2
- 26/05 → Support challenge 2 and pre-test challenge
- 27/05 → NO CLASS!
- 02/06 → NO CLASS!
- 03/06 → Second challenge



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**Thank you for your attention!**